

Project-Q Workbook - 6

30-Day Quantum Computing Challenge

Day - 06

Single-Qubit Quantum Gates

X • Y • Z • H Gates

Quick Recap (From Day 1–5)

Before we learn quantum gates, let's quickly recall what we already know.

- A **qubit** is the basic unit of quantum information.
- A qubit is represented as a **quantum state**:
$$\begin{bmatrix} |\psi\rangle = \alpha|0\rangle + \beta|1\rangle \end{bmatrix}$$
- The **Bloch Sphere** helps us visualize a single qubit.
- **Superposition** allows a qubit to exist as a combination of basis states.
- **Probability amplitudes** determine measurement probabilities.
- **Measurement** converts a quantum state into a classical outcome.
- After measurement, the qubit undergoes **state collapse**.

Today's Question:

If we know what a qubit is, how do we change its state?

The answer is: **Quantum Gates**

Why Do We Need Quantum Gates?

Imagine opening a text editor where you can only read text but cannot type, delete, or edit anything. That software would be useless. Similarly, a quantum computer that cannot modify qubits cannot perform computations. Quantum gates are the building blocks that **change the**

state of qubits. Just as classical computers use logic gates (AND, OR, NOT), quantum computers use quantum gates to perform computations.

Classical Gates vs Quantum Gates

Classical Gates	Quantum Gates
Operate on bits	Operate on qubits
Perform logical operations	Perform state transformations
Input is always 0 or 1	Input can be any valid quantum state
Example: NOT gate	Example: X Gate

Unlike classical gates, quantum gates are **reversible**. This means every gate has an inverse operation, allowing quantum computations to preserve information.

Understanding Gates Using the Bloch Sphere

Recall from Day 3 that a qubit can be represented as a point on the Bloch Sphere.

Instead of thinking of gates as "changing 0 to 1", think of them as **rotating the qubit's state** on the sphere.

Different gates perform different rotations, leading to different quantum states.

1. X Gate (Quantum NOT Gate)

What is the X Gate?

The **X Gate** is the quantum equivalent of the classical NOT gate.

It swaps the computational basis states:

$$|0\rangle \xrightarrow{X} |1\rangle$$

$$|1\rangle \xrightarrow{X} |0\rangle$$

For basic states, it behaves exactly like a NOT operation.

Matrix Representation

```
[  
X=  
\begin{bmatrix}  
0&1\  
1&0  
\end{bmatrix}  
]
```

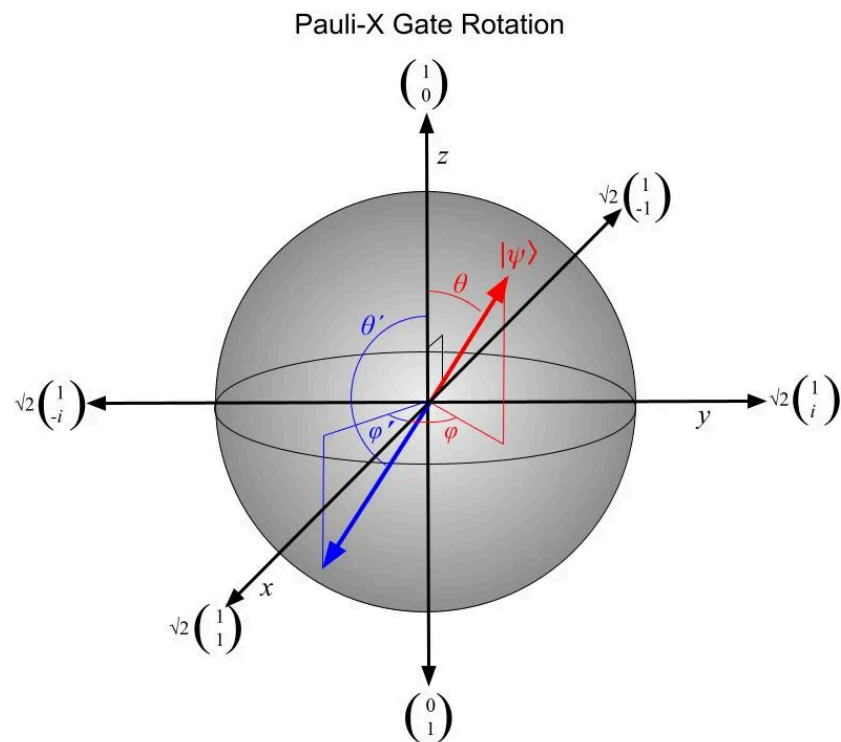
You don't need to memorize the matrix yet. Just remember that it swaps the amplitudes of $|0\rangle$ and $|1\rangle$.

Bloch Sphere View

The X Gate rotates the qubit by **180° about the X-axis**.

Imagine flipping the Bloch Sphere upside down around the horizontal X-axis.

North Pole ($|0\rangle$) becomes South Pole ($|1\rangle$), and vice versa.



Example

Initial State: $|0\rangle$

Apply X Gate:

$|0\rangle$
 $\downarrow$
X
 $\downarrow$
 $|1\rangle$

If the input is: $|1\rangle$

Output: $|0\rangle$

Remember This

- Acts like a NOT gate for basis states.
 - Swaps ($|0\rangle$) and ($|1\rangle$).
 - Rotates the Bloch Sphere about the X-axis.
 - Works on **any** quantum state, not just basis states.
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2. Y Gate

What is the Y Gate?

The **Y Gate** performs two actions simultaneously:

- Flips the qubit.
- Changes its phase.

Because it changes both the state and the phase, it is slightly more complex than the X Gate.

Matrix Representation

[
Y=

```

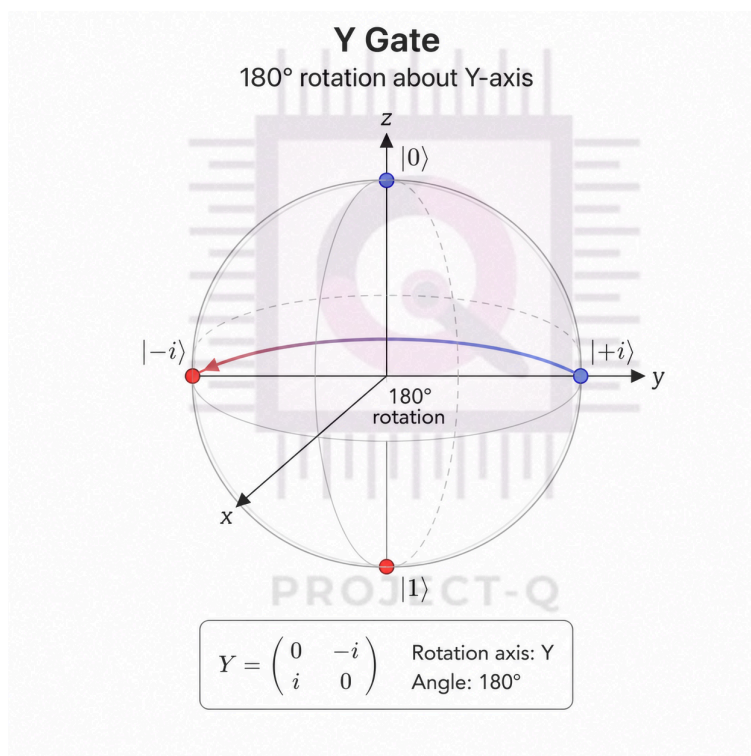
\begin{bmatrix}
0&-i\\
i&0
\end{bmatrix}

```

The presence of (i) (the imaginary unit) indicates that this gate also modifies the phase.

Bloch Sphere View

The Y Gate rotates the qubit by **180° about the Y-axis**.



Example

$|0\rangle$
 $\downarrow$
 Y
 $\downarrow$
 $i|1\rangle$

The extra (i) represents a phase change.

Remember This

- Flips the qubit.
 - Introduces a phase change.
 - Rotates around the Y-axis.
 - Less commonly used directly than X or H but important in many quantum operations.
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3. Z Gate

What is the Z Gate?

Unlike the X Gate, the Z Gate does **not** swap $|0\rangle$ and $|1\rangle$.

Instead, it changes the **phase** of the $|1\rangle$ state.

$$|0\rangle \xrightarrow{Z} |0\rangle$$

$$|1\rangle \xrightarrow{Z} -|1\rangle$$

Notice that only the sign changes.

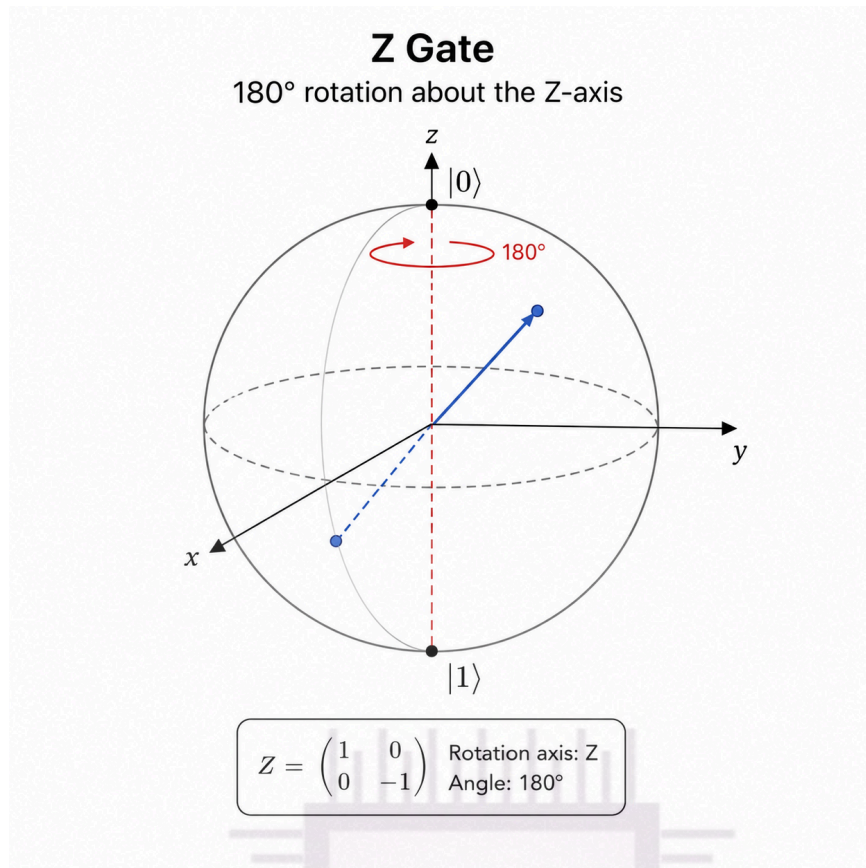
Why Does This Matter?

Although the measurement probabilities remain the same, the changed phase affects how the qubit behaves when combined with other gates.

This is one of the reasons quantum algorithms can exploit **interference**.

Bloch Sphere View

The Z Gate rotates the qubit by **180° about the Z-axis**.



Remember This

- Does not change measurement probabilities immediately.
 - Changes the phase of the quantum state.
 - Essential for creating interference in quantum algorithms.
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4. Hadamard (H) Gate

Why Is the H Gate Special?

The Hadamard Gate is one of the most important gates in quantum computing.

Its primary purpose is to create **equal superposition**.

Action of the H Gate

Starting from:

$|0\rangle$

Applying H gives:

$$\left[\frac{|0\rangle + |1\rangle}{\sqrt{2}} \right]$$

Starting from:

$|1\rangle$

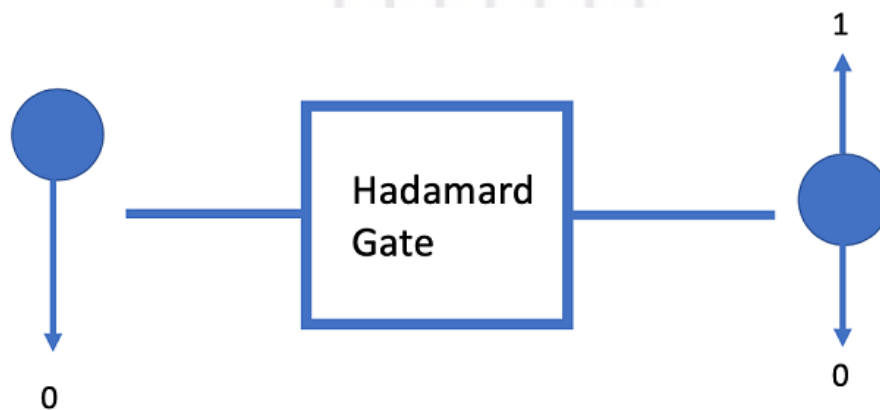
Applying H gives:

$$\left[\frac{|0\rangle - |1\rangle}{\sqrt{2}} \right]$$

Notice that the second state has a different **phase**.

Bloch Sphere View

Instead of simply flipping the qubit, the H Gate moves the state from the Z-axis toward the X-axis, creating an equal superposition.



Why Is It Important?

Many famous quantum algorithms begin by applying the H Gate because it prepares qubits to explore multiple computational possibilities.

Examples include:

- Grover's Search Algorithm
 - Deutsch-Jozsa Algorithm
 - Shor's Algorithm
 - Quantum Fourier Transform
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Remember This

- Creates equal superposition from basis states.
 - One of the most frequently used quantum gates.
 - Often the first gate is applied in a quantum algorithm.
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Comparing the Four Gates

Gate	Main Action	Bloch Sphere Rotation	Changes Phase?
X	Flips basis states	180° about X-axis	No
Y	Flip + Phase	180° about Y-axis	Yes
Z	Phase Flip	180° about Z-axis	Yes
H	Creates Superposition	Moves state toward X-basis	Yes

Common Beginner Mistakes

"The X Gate is always the same as a NOT gate."

✓ It behaves like NOT only for basis states. For arbitrary quantum states, it transforms the amplitudes.

"The Z Gate does nothing because the measurement doesn't change."

✓ The Z Gate changes the phase, which affects later operations.

"The H Gate gives both 0 and 1 as outputs."

✓ It prepares the qubit in a superposition. A measurement still yields only one classical outcome.

End-of-Day Summary

Today You Learned

- Why quantum gates are needed.
- Difference between classical and quantum gates.
- X Gate (Bit Flip).
- Y Gate (Bit Flip + Phase Change).
- Z Gate (Phase Flip).
- Hadamard Gate (Superposition Creator).
- Bloch Sphere interpretation of each gate.
- Common misconceptions.

Top Verified Learning Resources

Recommended Video

Single-Qubit Gates Explained

[Single-Qubit Gates \(YouTube\)](#)

This video provides an intuitive explanation of:

- X Gate
- Y Gate
- Z Gate
- Hadamard Gate
- Bloch Sphere rotations

It's an excellent companion to the workbook after completing the theory.